José María Rodríguez García

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PROFILE

AI Software Engineer who builds AI products end to end: the agents, the interfaces and the cloud they run on. Currently lead developer of a digital signage platform driven by AI agents and contributor to a robotics research project at NTT DATA's Innovation Center in Barcelona.

EXPERIENCE

NTT DATA Europe & Latam — Barcelona, Spain
AI Software Engineer, Innovation Center — Dec 2025 – Present

- Lead developer of Agentic Space Demo, a digital signage platform driven by AI agents (see Projects).
- Co-developed Stand Allocation Optimizer, an airport operations proof of concept (see Projects).
- Contributor to a robotics research project with a mobile manipulator.
- Built real-time state synchronisation over WebSockets on FastAPI, delivering updates to every connected client in under 100 ms.
- Hackathon semifinalist with an AI research assistant built in 48 hours: document ingestion, retrieval of the relevant passages and answers generated by an LLM.

Revolution Computer — Barcelona, Spain
Computer Technician — Mar 2019 – Sep 2019

- Diagnosed and repaired hardware and software issues; assembled and configured PCs from scratch.
- Provided in-person technical support to customers.

SELECTED PROJECTS

Agentic Space Demo, *Lead developer* — Dec 2025 – Present
Python, FastAPI, React, Azure OpenAI, AI agents, MCP, embeddings, Cosmos DB, Terraform

- Automated the preparation of client visits, previously a manual process of choosing videos and slides and configuring each screen one by one.
- Ingestion pipeline that transcribes video (Whisper), describes images, extracts text from PDF and PowerPoint, and indexes it all with embeddings for semantic search.
- Orchestrator with specialist agents that scripts the visit and assigns content to each screen by size and orientation; a conversational assistant asks three one-click questions and changes nothing until the plan is confirmed.
- Interactive floor plan with live screen status and the visit route; per-user AI budgets with limits and approvals.

Stand Allocation Optimizer, *Built with a teammate* — Jun 2026 – Sep 2026
React, TypeScript, Three.js, Mapbox, D3, FastAPI, OR-Tools, Azure, Docker, Terraform

- Proof of concept that assigns aircraft to stands and coordinates ground movement for a full day of flights at a large airport, detecting, explaining and resolving conflicts.
- Geospatial pipeline on OpenStreetMap with Dijkstra taxi routes; stand optimiser with OR-Tools (CP-SAT) that removes conflicts and reduces the use of remote stands.
- Taxiway and runway conflict detection, plus an optimiser that sequences aircraft with bounded holds.
- Operations console in React with a satellite map (Mapbox), 3D view (Three.js) and Gantt chart (D3) on a shared clock; deployed on Azure with Docker and Terraform.

EDUCATION

CEPI-BASE — Barcelona, Spain
Higher Vocational Degree (CFGS) in Application Development with Python — Sep 2021 – Oct 2024

Institut Pedralbes — Barcelona, Spain
Intermediate Vocational Degree (CFGM) in Microcomputer Systems and Networks — Jun 2018 – Sep 2020

SKILLS

Programming: Python, TypeScript, JavaScript, SQL
AI: Azure OpenAI, AI agents and orchestration, MCP, embeddings and semantic search, RAG, Whisper
Frontend: React, Three.js, D3, Mapbox, Tailwind CSS, HTML, CSS
Backend: FastAPI, Node.js, REST APIs, WebSockets, WebRTC, OR-Tools (CP-SAT)
Data: PostgreSQL, MongoDB, Cosmos DB, MySQL
Cloud & DevOps: Azure, Docker, Terraform, Git
Languages: Spanish (native), Catalan (native), English (B2, professional working proficiency)